

Maxwell Equations Conservation Laws

PHYS505 – Electromagnetic Theory

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Maxwell equations

- Ampère's Law modified

$$\vec{\nabla} \times \vec{H} = \vec{J} + \partial_t \vec{D}$$

- to include the displacement current
- satisfies the continuity equation

$$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{H}) = 0$$

$$\vec{\nabla} \cdot \vec{J} + \partial_t \rho = 0$$

- resulting in the final form

$$\vec{\nabla} \cdot \vec{D} = \rho$$

$$\vec{\nabla} \times \vec{H} = \vec{J} + \partial_t \vec{D}$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \times \vec{E} = -\partial_t \vec{B}$$

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- 4 coupled 1st-order partial differential eqns in 16 components
 - 4 vector fields $\vec{E}, \vec{D}, \vec{B}, \vec{H}$
 - charge and current densities $\rho, \vec{J}$
- plus constitutive relations for $\vec{E} \Leftrightarrow \vec{D}, \vec{B} \Leftrightarrow \vec{H}$
- Often more convenient to solve fewer (2nd-order) eqns in the potentials
- Vector potential

$$\vec{B} = \vec{\nabla} \times \vec{A}$$

- Scalar potential

$$\vec{E} + \partial_t \vec{A} = -\vec{\nabla} \Phi \rightarrow \vec{E} = -\vec{\nabla} \Phi - \partial_t \vec{A}$$

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Vacuum – Lorenz condition

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Gauge transformations – Lorenz gauge

$$\begin{array}{ll} \vec{A} \rightarrow \vec{A} + \vec{\nabla}\Lambda & \vec{B} \rightarrow \vec{B} \\ \Phi \rightarrow \Phi - \partial_t\Lambda & \vec{E} \rightarrow \vec{E} \end{array} \qquad \vec{\nabla} \cdot \vec{A} + \frac{1}{c^2} \partial_t \Phi = 0$$

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- Lorenz condition decouples the wave eqns for the potentials
- For potentials satisfying the Lorenz condition (aka Lorenz gauge), fields invariant under gauge transformations with Λ that obeys

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Gauge transformations – Coulomb gauge

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$$\nabla^2 \Phi = -\frac{\rho}{\epsilon_0}$$
$$\nabla^2 \vec{A} - \frac{1}{c^2} \partial_t^2 \vec{A} = -\mu_0 \vec{J} + \frac{1}{c^2} \vec{\nabla} (\partial_t \Phi)$$

- aka radiation gauge, transverse gauge
- solution for Φ is the instantaneous Coulomb potential due to ρ

$$\Phi(\vec{x}, t) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\vec{x}', t)}{|\vec{x} - \vec{x}'|} d^3x'$$

- solution for $\vec{A}$ involves the transverse part of $\vec{J}$

$$\vec{J}_t(\vec{x}, t) = \frac{1}{4\pi} \vec{\nabla} \times \vec{\nabla} \times \int \frac{\vec{J}(\vec{x}', t)}{|\vec{x} - \vec{x}'|} d^3x'$$

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Green functions for the wave equation

- Looking for solution to

$$\nabla^2 \psi - \frac{1}{c^2} \partial_t^2 \psi = -4\pi f(\vec{x}, t)$$

with given source distribution $f(\vec{x}, t)$

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- In terms of Fourier transforms

$$\psi(\vec{x}, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \psi(\vec{x}, \omega) e^{-i\omega t} d\omega$$

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- the (Helmholtz) wave equation reads (with $k = \omega/c$)

$$(\nabla^2 + k^2) \psi(\vec{x}, \omega) = -4\pi f(\vec{x}, \omega)$$

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$$(\nabla^2 + k^2) G_k(\vec{x}, \vec{x}') = -4\pi \delta(\vec{x} - \vec{x}')$$

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- If no boundaries, $G_k = G_k(R)$ with $\vec{R} = \vec{x} - \vec{x}'$

$$\frac{1}{R} \frac{d^2}{dR^2} [R G_k(R)] + k^2 G_k(R) = -4\pi \delta(\vec{R})$$

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- and is of the form

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- Going back to the time domain

$$\left(\nabla^2 - \frac{1}{c^2} \partial_t^2 \right) G^{(\pm)}(\vec{x}, t; \vec{x}', t') = -4\pi \delta(\vec{x} - \vec{x}') \delta(t - t')$$

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- the solutions are

$$G^{(\pm)}(R, \tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{e^{\pm i k R}}{R} e^{-i \omega \tau} d\omega, \quad \tau = t - t'$$

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- the solutions are

$$G^{(\pm)}(R, \tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{e^{\pm i k R}}{R} e^{-i \omega \tau} d\omega, \quad \tau = t - t'$$

- For a non-dispersive medium $k = \omega/c$

$$G^{(\pm)}(R, \tau) = \frac{1}{R} \delta\left(\tau \mp \frac{R}{c}\right)$$

Green functions for the wave equation

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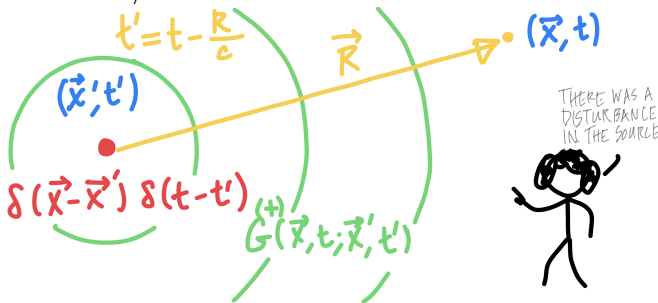
- $G^{(+)}$ is called the *retarded Green function* and represents causal behavior: the effect of a disturbance from a source at distance R that arrives at time t is from *retarded time* $t' = t - R/c$

Green functions for the wave equation

- Going back to the time domain

$$\left(\nabla^2 - \frac{1}{c^2} \partial_t^2 \right) G^{(\pm)}(\vec{x}, t; \vec{x}', t') = -4\pi \delta(\vec{x} - \vec{x}') \delta(t - t')$$

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- $G^{(-)}$ is called the *advanced Green function*

Green functions for the wave equation

- Looking for solution to

$$\nabla^2 \psi - \frac{1}{c^2} \partial_t^2 \psi = -4\pi f(\vec{x}, t)$$

with given source distribution $f(\vec{x}, t)$

- $G^{(-)}$ is called the *advanced Green function*
- Particular solution using Green functions

$$\psi^{(\pm)}(\vec{x}, t) = \int \int G^{(\pm)}(\vec{x}, t; \vec{x}', t') f(\vec{x}', t') d^3x' dt'$$

Green functions for the wave equation

$$\psi^{(\pm)}(\vec{x}, t) = \int \int G^{(\pm)}(\vec{x}, t; \vec{x}', t') f(\vec{x}', t') d^3x' dt'$$

Assume source $f(\vec{x}', t')$ localized in space and time

- With incoming wave $\psi_{in}(\vec{x}, t)$ at $t \rightarrow -\infty$ satisfying homogeneous wave eqn

$$\psi(\vec{x}, t) = \psi_{in}(\vec{x}, t) + \int \int G^{(+)}(\vec{x}, t; \vec{x}', t') f(\vec{x}', t') d^3x' dt'$$

$$\psi_{in} = 0 \rightarrow \psi(\vec{x}, t) = \int \frac{[f(\vec{x}', t')]_{ret}}{|\vec{x} - \vec{x}'|} d^3x'$$

with the source contribution evaluated at retarded time
 $t' = t - |\vec{x} - \vec{x}'|/c$

- With outgoing wave $\psi_{out}(\vec{x}, t)$ at $t \rightarrow +\infty$

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Retarded solutions for the fields

$$\Phi(\vec{x}, t) = \frac{1}{4\pi\epsilon_0} \int d^3x' \frac{1}{R} [\rho(\vec{x}', t')]_{ret}$$

$$\vec{A}(\vec{x}, t) = \frac{\mu_0}{4\pi\epsilon_0} \int d^3x' \frac{1}{R} [\vec{J}(\vec{x}', t')]_{ret}$$

$$\vec{R} = \vec{x} - \vec{x}', \quad R = |\vec{R}|$$

- Fields can be computed from these using

$$\vec{B} = \vec{\nabla} \times \vec{A}, \quad \vec{E} = -\vec{\nabla}\Phi - \partial_t \vec{A}$$

- Sometimes more convenient to solve for the fields directly
– in free space, we have

$$\nabla^2 \vec{E} - \frac{1}{c^2} \partial_t^2 \vec{E} = \frac{1}{\epsilon_0} \left(\vec{\nabla} \rho + \frac{1}{c^2} \partial_t \vec{J} \right)$$

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- General solution

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- Formal solutions

$$\vec{E}(\vec{x}, t) = -\frac{1}{4\pi\epsilon_0} \int d^3x' \frac{1}{R} \left[\vec{\nabla}' \rho(\vec{x}', t') + \frac{1}{c^2} \partial_{t'} \vec{J}(\vec{x}', t') \right]_{ret}$$

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$$\left[\vec{\nabla}' \rho \right]_{ret} = \vec{\nabla}' [\rho]_{ret} - \frac{\hat{\mathbf{R}}}{c} [\partial_{t'} \rho]_{ret}$$

$$\left[\vec{\nabla}' \times \vec{J} \right]_{ret} = \vec{\nabla}' \times [\vec{J}]_{ret} + \left[\partial_{t'} \vec{J} \right]_{ret} \times \frac{\hat{\mathbf{R}}}{c}$$

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- Jefimenko formulas for generalization of Coulomb and Biot-Savart laws

$$\begin{aligned}\vec{E}(\vec{x}, t) &= \frac{1}{4\pi\epsilon_0} \int d^3x' \left\{ \frac{[\rho]_{ret}}{R^2} \hat{\mathbf{R}} + \frac{[\partial_{t'} \rho]_{ret}}{cR} \hat{\mathbf{R}} - \frac{1}{c^2 R} \left[\partial_{t'} \vec{J}\right]_{ret} \right\} \\ \vec{B}(\vec{x}, t) &= \frac{\mu_0}{4\pi} \int d^3x' \left\{ \frac{1}{R^2} [\vec{J}]_{ret} \times \hat{\mathbf{R}} + \frac{1}{cR} \left[\partial_{t'} \vec{J}\right]_{ret} \times \hat{\mathbf{R}} \right\}\end{aligned}$$

Retarded solutions for the fields

- Heaviside-Feynman formulas for point charge with trajectory $\vec{r}_0(t)$ and velocity $\vec{v}$, with $\kappa = 1 - \vec{v} \cdot \hat{\mathbf{R}}/c$

$$\rho(\vec{x}', t') = q \delta(\vec{x}' - \vec{r}_0(t')), \quad \vec{J}(\vec{x}', t') = \rho(\vec{x}', t') \vec{v}(t')$$

$$\begin{aligned} \vec{E}(\vec{x}, t) &= \frac{q}{4\pi\epsilon_0} \left\{ \left[\frac{\hat{\mathbf{R}}}{\kappa R^2} \right]_{ret} + \frac{\partial}{c\partial t} \left[\frac{\hat{\mathbf{R}}}{\kappa R} \right]_{ret} - \frac{\partial}{c^2\partial t} \left[\frac{\vec{v}}{\kappa R} \right]_{ret} \right\} \\ &= \frac{q}{4\pi\epsilon_0} \left\{ \left[\frac{\hat{\mathbf{R}}}{R^2} \right]_{ret} + \frac{[R]_{ret}}{c} \frac{\partial}{\partial t} \left[\frac{\hat{\mathbf{R}}}{R^2} \right]_{ret} + \frac{\partial^2}{c^2\partial t^2} [\hat{\mathbf{R}}]_{ret} \right\} \end{aligned}$$

$$\begin{aligned} \vec{B}(\vec{x}, t) &= \frac{\mu_0 q}{4\pi} \left\{ \left[\frac{\vec{v} \times \hat{\mathbf{R}}}{\kappa R^2} \right]_{ret} + \frac{\partial}{c\partial t} \left[\frac{\vec{v} \times \hat{\mathbf{R}}}{\kappa R} \right]_{ret} \right\} \\ &= \frac{\mu_0 q}{4\pi} \left\{ \left[\frac{\vec{v} \times \hat{\mathbf{R}}}{\kappa^2 R^2} \right]_{ret} + \frac{1}{c[R]_{ret}} \frac{\partial}{\partial t} \left[\frac{\vec{v} \times \hat{\mathbf{R}}}{\kappa} \right]_{ret} \right\} \end{aligned}$$

Macroscopic vs microscopic

- Maxwell eqns describing macroscopic fields

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \times \vec{E} + \partial_t \vec{B} = 0$$

$$\vec{\nabla} \cdot \vec{D} = \rho$$

$$\vec{\nabla} \times \vec{H} - \partial_t \vec{D} = \vec{J}$$

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P}$$

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- using spatial averaging over a scale of order $L_0 \approx 10 \text{ nm}$

$$\langle F(\vec{x}, t) \rangle = \int d^3x' f(\vec{x}') F(\vec{x} - \vec{x}', t)$$

with some test function f , eg.

$$f(\vec{x}) = \frac{1 - \Theta(r - R)}{\frac{4}{3}\pi R^3}, \quad f(\vec{x}) = \frac{e^{-\frac{r^2}{R^2}}}{(\pi R^2)^{3/2}}$$

$$\vec{E}(\vec{x}, t) = \langle \vec{e}(\vec{x}, t) \rangle, \quad \vec{B}(\vec{x}, t) = \langle \vec{b}(\vec{x}, t) \rangle$$

Macroscopic vs microscopic

- so that

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Poynting's theorem – energy conservation

- Work done on point charge by external fields $\vec{E}, \vec{B}$ is $q\vec{v} \cdot \vec{E}$
- Electromagnetic energy used in a region with continuous charge, current, at a rate of

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$$u = \frac{1}{2} (\vec{E} \cdot \vec{D} + \vec{B} \cdot \vec{H})$$

is also valid for the time-dependent case

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- for a linear medium with negligible dispersion and loss, the rate of change is given by

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converted into kinetic (mechanical, thermal) energy

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- Conservation of energy for a linear medium

$$\partial_t u + \vec{\nabla} \cdot \vec{S} = -\vec{J} \cdot \vec{E}, \quad \vec{S} = \vec{E} \times \vec{H}$$

in terms of the *Poynting vector* $\vec{S}$ representing energy flux

Momentum conservation

- Force $\vec{F}$ on a charged particle due to external fields $\vec{E}, \vec{B}$

$$\vec{F} = q \left(\vec{E} + \vec{v} \times \vec{B} \right)$$

- Rate of change of total linear momentum for charge distribution

$$\frac{d\vec{P}_{mech}}{dt} = \int_V \left(\rho \vec{E} + \vec{J} \times \vec{B} \right) d^3x$$

- In linear media, this leads to

$$\frac{d\vec{P}_{mech}}{dt} + \frac{d\vec{P}_{field}}{dt} = \oint_S \sum_{\beta} T_{\alpha\beta} n_{\beta} da$$

in terms of the electromagnetic momentum density $\vec{g}$ and
Maxwell stress tensor $T_{\alpha\beta}$

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Linear dispersive media with losses

- Fourier decomposition of fields

$$\begin{aligned}\vec{E}(\vec{x}, t) &= \int_{-\infty}^{\infty} d\omega \vec{E}(\vec{x}, \omega) e^{-i\omega t} & \vec{D}(\vec{x}, t) &= \int_{-\infty}^{\infty} d\omega \vec{D}(\vec{x}, \omega) e^{-i\omega t} \\ \vec{B}(\vec{x}, t) &= \int_{-\infty}^{\infty} d\omega \vec{B}(\vec{x}, \omega) e^{-i\omega t} & \vec{H}(\vec{x}, t) &= \int_{-\infty}^{\infty} d\omega \vec{H}(\vec{x}, \omega) e^{-i\omega t}\end{aligned}$$

- For linear, isotropic media

$$\vec{D}(\vec{x}, \omega) = \epsilon(\omega) \vec{E}(\vec{x}, \omega), \quad \vec{B}(\vec{x}, \omega) = \mu(\omega) \vec{H}(\vec{x}, \omega)$$

- Time-domain fields are real

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Linear dispersive media with losses

- Dispersion (frequency-dependent ϵ, μ) means in

$$\int_V \vec{J} \cdot \vec{E} d^3x = - \int_V \left[\vec{\nabla} \cdot (\vec{E} \times \vec{H}) + \vec{E} \cdot \partial_t \vec{D} + \vec{H} \cdot \partial_t \vec{B} \right] d^3x$$

the ∂_t terms need special consideration

- For the $\vec{E}, \vec{H}$ -fields in a narrow range of frequencies around ω_0 , Poynting's theorem becomes

$$u_{\text{eff}} = \frac{1}{2} \left\{ \text{Re} \left[\frac{d(\omega\epsilon)}{d\omega}(\omega_0) \right] \langle |\vec{E}(\vec{x}, t)|^2 \rangle + \text{Re} \left[\frac{d(\omega\mu)}{d\omega}(\omega_0) \right] \langle |\vec{H}(\vec{x}, t)|^2 \rangle \right\}$$

$$\begin{aligned} \partial_t u_{\text{eff}} + \vec{\nabla} \cdot \vec{S} = & -\vec{J} \cdot \vec{E} \\ & - \omega_0 \left\{ \text{Im} [\epsilon(\omega_0)] \langle |\vec{E}(\vec{x}, t)|^2 \rangle + \text{Im} [\mu(\omega_0)] \langle |\vec{H}(\vec{x}, t)|^2 \rangle \right\} \end{aligned}$$

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Harmonic fields

- Assuming harmonic time dependence for fields

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where $\vec{E}(\vec{x})$ is in general complex

- Maxwell eqns in the complex fields

$$\begin{aligned} \vec{\nabla} \cdot \vec{B} &= 0 & \vec{\nabla} \times \vec{E} - i\omega \vec{B} &= 0 \\ \vec{\nabla} \cdot \vec{D} &= \rho & \vec{\nabla} \times \vec{H} + i\omega \vec{D} &= \vec{J} \end{aligned}$$

- Poynting vector, energy densities

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- $\det(a) = 1$ represents a proper rotation that can be built up in infinitesimal steps while $\det(a) = -1$ includes a reflection
- Quantities that transform this way are called *rank-one tensors* or *vectors* – particle position $\vec{x}_i$, velocity $\vec{v}_i$, momentum $\vec{p}_i$

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- Second-rank tensors, or simply *tensors* transform according to

$$B'_{\alpha\beta} = \sum_{\gamma,\delta} a_{\alpha\gamma} a_{\beta\delta} B_{\gamma\delta}$$

and include the Maxwell stress tensor

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- The cross product transforms as a vector under proper rotations

$$\vec{A} = \vec{B} \times \vec{C}, \quad A_\alpha = \sum_{\beta,\gamma} \epsilon_{\alpha\beta\gamma} B_\beta C_\gamma$$

- Reflection in a plane means reversing the sign of the coordinates along the $\perp$ to the plane, eg. for $x - z$ plane

$$\vec{x} = (x, y, z) \rightarrow \vec{x}' = (x', y', z') = (x, -y, z)$$

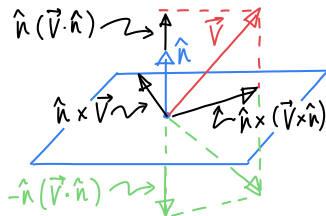
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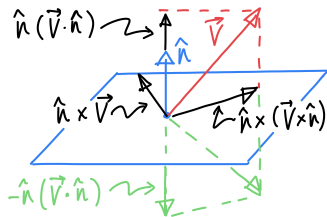
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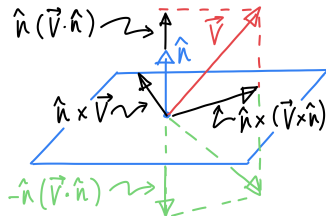
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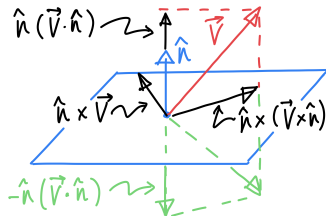
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- Caveat lector (vector?) – the presence/absence of a vector sign/font for a symbol doesn't indicate which type of scalar/vector it transforms as

Transformations – time reversal

- At the microscopic level, classical physics is invariant under $t \leftrightarrow -t$
- meaning the equations expressing physical laws retain their form with appropriate transformation of the variables
- For basic kinematic variables

$$\vec{x} \rightarrow \vec{x}' = \vec{x}, \quad \vec{v} = \frac{d}{dt}\vec{x} \rightarrow \vec{v}' = -\vec{v}, \quad \vec{p} \rightarrow -\vec{p}$$

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$$\vec{x} \rightarrow \vec{x}' = \vec{x}, \quad \vec{v} = \frac{d}{dt}\vec{x} \rightarrow \vec{v}' = -\vec{v}, \quad \vec{p} \rightarrow -\vec{p}$$

- For particle in external potential

$$\frac{d}{dt}\vec{p} = -\vec{\nabla} U(\vec{x})$$

- Running the clock in reverse from a final state of the system, equivalent to reversing all the momenta, gives the same initial state

Transformations – electromagnetic quantities

	Rotation (rank)	Inversion	Time reversal
q	0	even (scalar)	even
ρ	0	even (scalar)	even
$\vec{J}$	1	odd (vector)	odd
$\vec{E}$	1	odd (vector)	even
$\vec{B}$	1	even (pseudovector)	odd
$\vec{S}$	1	odd (vector)	odd
$T_{\alpha\beta}$	2	even (tensor)	even

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

$$\vec{\nabla} \times \vec{E} = -\partial_t \vec{B}$$

$$\vec{\nabla} \times \vec{B} = \frac{1}{c^2} \partial_t \vec{E} + \mu_0 \vec{J}$$

$$\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B}$$

$$T_{\alpha\beta} = \epsilon_0 \left[E_\alpha E_\beta + c^2 B_\alpha B_\beta - \frac{\delta_{\alpha\beta}}{2} (E^2 + c^2 B^2) \right]$$

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Magnetic monopoles?

- Starting from Maxwell equations
- with electric charge and current density
- What if we also had magnetic charge and current density?
- with the corresponding conservation/continuity eqn

$$\vec{\nabla} \cdot \vec{D} = \rho$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \times \vec{H} = \partial_t \vec{D} + \vec{J}$$

$$-\vec{\nabla} \times \vec{E} = \partial_t \vec{B}$$

Magnetic monopoles?

- Starting from Maxwell equations
- with **electric** charge and current density
- What if we also had magnetic charge and current density?
- with the corresponding conservation/continuity eqn

$$\vec{\nabla} \cdot \vec{D} = \rho_e$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \times \vec{H} = \partial_t \vec{D} + \vec{J}_e$$

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$$\vec{\nabla} \cdot \vec{D} = \rho_e$$

$$\vec{\nabla} \cdot \vec{B} = \rho_m$$

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Magnetic monopoles?

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$$\begin{array}{lll} \vec{\nabla} \cdot \vec{D} = \rho_e & \vec{\nabla} \times \vec{H} = \partial_t \vec{D} + \vec{J}_e & 0 = \partial_t \rho_e + \vec{\nabla} \cdot \vec{J}_e \\ \vec{\nabla} \cdot \vec{B} = \rho_m & -\vec{\nabla} \times \vec{E} = \partial_t \vec{B} + \vec{J}_m & 0 = \partial_t \rho_m + \vec{\nabla} \cdot \vec{J}_m \end{array}$$

Magnetic monopoles?

- Starting from Maxwell equations
- with electric charge and current density
- What if we also had magnetic charge and current density?
- with the corresponding conservation/continuity eqn
- rewritten in matrix form

$$\begin{aligned}\vec{\nabla} \cdot \vec{D} &= \rho_e & \vec{\nabla} \times \vec{H} &= \partial_t \vec{D} + \vec{J}_e \\ \vec{\nabla} \cdot \vec{B} &= \rho_m & -\vec{\nabla} \times \vec{E} &= \partial_t \vec{B} + \vec{J}_m\end{aligned}$$

$$\vec{\nabla} \cdot \begin{pmatrix} \vec{D} \\ \vec{B} \end{pmatrix} = \begin{pmatrix} \rho_e \\ \rho_m \end{pmatrix}, \quad \vec{\nabla} \times \begin{pmatrix} \vec{H} \\ -\vec{E} \end{pmatrix} = \partial_t \begin{pmatrix} \vec{D} \\ \vec{B} \end{pmatrix} + \vec{\nabla} \cdot \begin{pmatrix} \vec{J}_e \\ \vec{J}_m \end{pmatrix}$$

Magnetic monopoles?

- Starting from Maxwell equations
- with electric charge and current density
- What if we also had magnetic charge and current density?
- with the corresponding conservation/continuity eqn
- rewritten in matrix form with matched dimensions
($Z_0 = \sqrt{\mu_0/\epsilon_0} = \mu_0 c$)

$$\vec{\nabla} \cdot \vec{D} = \rho_e$$

$$\vec{\nabla} \times \vec{H} = \partial_t \vec{D} + \vec{J}_e$$

$$\vec{\nabla} \cdot \vec{B} = \rho_m$$

$$-\vec{\nabla} \times \vec{E} = \partial_t \vec{B} + \vec{J}_m$$

$$\vec{\nabla} \cdot \begin{pmatrix} Z_0 \vec{D} \\ \vec{B} \end{pmatrix} = \begin{pmatrix} Z_0 \rho_e \\ \rho_m \end{pmatrix}, \quad \vec{\nabla} \times \begin{pmatrix} Z_0 \vec{H} \\ -\vec{E} \end{pmatrix} = \partial_t \begin{pmatrix} Z_0 \vec{D} \\ \vec{B} \end{pmatrix} + \vec{\nabla} \cdot \begin{pmatrix} Z_0 \vec{J}_e \\ \vec{J}_m \end{pmatrix}$$

Duality transformations

- Symmetrized Maxwell eqns with magnetic charge/current

$$\vec{\nabla} \cdot \begin{pmatrix} Z_0 \vec{D} \\ \vec{B} \end{pmatrix} = \begin{pmatrix} Z_0 \rho_e \\ \rho_m \end{pmatrix}, \quad \vec{\nabla} \times \begin{pmatrix} -\vec{E} \\ Z_0 \vec{H} \end{pmatrix} = \partial_t \begin{pmatrix} \vec{B} \\ Z_0 \vec{D} \end{pmatrix} + \vec{\nabla} \cdot \begin{pmatrix} \vec{J}_m \\ Z_0 \vec{J}_e \end{pmatrix}$$

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- Consider the proper orthogonal transformation $R(\xi)$

$$R(\xi) = \begin{pmatrix} \cos \xi & \sin \xi \\ -\sin \xi & \cos \xi \end{pmatrix}, \quad R(\xi)^T = R(\xi)^{-1} = R(-\xi)$$

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applied to "rotate" between electric/magnetic fields

$$\begin{pmatrix} \vec{E} \\ Z_0 \vec{H} \end{pmatrix} = \begin{pmatrix} \cos \xi & \sin \xi \\ -\sin \xi & \cos \xi \end{pmatrix} \begin{pmatrix} \vec{E}' \\ Z_0 \vec{H}' \end{pmatrix}$$
$$\begin{pmatrix} Z_0 \vec{D} \\ \vec{B} \end{pmatrix} = \begin{pmatrix} \cos \xi & \sin \xi \\ -\sin \xi & \cos \xi \end{pmatrix} \begin{pmatrix} Z_0 \vec{D}' \\ \vec{B}' \end{pmatrix}$$

Duality transformations

- Symmetrized Maxwell eqns with magnetic charge/current

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$$\begin{pmatrix} Z_0 \vec{D} \\ \vec{B} \end{pmatrix} = \begin{pmatrix} \cos \xi & \sin \xi \\ -\sin \xi & \cos \xi \end{pmatrix} \begin{pmatrix} Z_0 \vec{D}' \\ \vec{B}' \end{pmatrix}$$

Duality transformations

- Symmetrized Maxwell eqns with magnetic charge/current

$$\vec{\nabla} \cdot \begin{pmatrix} Z_0 \vec{D} \\ \vec{B} \end{pmatrix} = \begin{pmatrix} Z_0 \rho_e \\ \rho_m \end{pmatrix}, \quad \vec{\nabla} \times \begin{pmatrix} -\vec{E} \\ Z_0 \vec{H} \end{pmatrix} = \partial_t \begin{pmatrix} \vec{B} \\ Z_0 \vec{D} \end{pmatrix} + \vec{\nabla} \cdot \begin{pmatrix} \vec{J}_m \\ Z_0 \vec{J}_e \end{pmatrix}$$

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$$\begin{pmatrix} -\vec{E} \\ Z_0 \vec{H} \end{pmatrix} = R(-\xi) \begin{pmatrix} -\vec{E}' \\ Z_0 \vec{H}' \end{pmatrix}$$
$$\begin{pmatrix} Z_0 \vec{D} \\ \vec{B} \end{pmatrix} = R(\xi) \begin{pmatrix} Z_0 \vec{D}' \\ \vec{B}' \end{pmatrix}$$

Duality transformations

- Symmetrized Maxwell eqns with magnetic charge/current

$$\vec{\nabla} \cdot \begin{pmatrix} Z_0 \vec{D} \\ \vec{B} \end{pmatrix} = \begin{pmatrix} Z_0 \rho_e \\ \rho_m \end{pmatrix}, \quad \vec{\nabla} \times \begin{pmatrix} -\vec{E} \\ Z_0 \vec{H} \end{pmatrix} = \partial_t \begin{pmatrix} \vec{B} \\ Z_0 \vec{D} \end{pmatrix} + \vec{\nabla} \cdot \begin{pmatrix} \vec{J}_m \\ Z_0 \vec{J}_e \end{pmatrix}$$

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$$\begin{aligned} \vec{\nabla} \times \begin{pmatrix} -\vec{E} \\ Z_0 \vec{H} \end{pmatrix} &= R(-\xi) \vec{\nabla} \times \begin{pmatrix} -\vec{E}' \\ Z_0 \vec{H}' \end{pmatrix} \\ \vec{\nabla} \cdot \begin{pmatrix} Z_0 \vec{D} \\ \vec{B} \end{pmatrix} &= R(\xi) \vec{\nabla} \cdot \begin{pmatrix} Z_0 \vec{D}' \\ \vec{B}' \end{pmatrix} = \begin{pmatrix} Z_0 \rho_e \\ \rho_m \end{pmatrix} \end{aligned}$$

Duality transformations

- Symmetrized Maxwell eqns with magnetic charge/current

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$$\vec{\nabla} \times \begin{pmatrix} -\vec{E} \\ Z_0 \vec{H} \end{pmatrix} = R(-\xi) \vec{\nabla} \times \begin{pmatrix} -\vec{E}' \\ Z_0 \vec{H}' \end{pmatrix} = \partial_t \begin{pmatrix} \vec{B} \\ Z_0 \vec{D} \end{pmatrix} + \vec{\nabla} \cdot \begin{pmatrix} \vec{J}_m \\ Z_0 \vec{J}_e \end{pmatrix}$$

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$$R(\xi) \vec{\nabla} \cdot \begin{pmatrix} Z_0 \vec{D}' \\ \vec{B}' \end{pmatrix} = \begin{pmatrix} Z_0 \rho_e \\ \rho_m \end{pmatrix}$$

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$$R(\xi) = \begin{pmatrix} \cos \xi & \sin \xi \\ -\sin \xi & \cos \xi \end{pmatrix}, \quad R(\xi)^T = R(\xi)^{-1} = R(-\xi)$$

and the reflection χ

$$\chi = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \chi \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} b \\ a \end{pmatrix}, \quad \chi^T = \chi^{-1} = \chi, \quad R\chi R = \chi$$

$$\vec{\nabla} \times \begin{pmatrix} -\vec{E}' \\ Z_0 \vec{H}' \end{pmatrix} = R(\xi) \partial_t [\chi \begin{pmatrix} Z_0 \vec{D} \\ \vec{B} \end{pmatrix}] + R(\xi) \vec{\nabla} \cdot \begin{pmatrix} \vec{J}_m \\ Z_0 \vec{J}_e \end{pmatrix}$$

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$$\vec{\nabla} \times \begin{pmatrix} -\vec{E}' \\ Z_0 \vec{H}' \end{pmatrix} = R(\xi) \partial_t [\chi R(\xi) \begin{pmatrix} Z_0 \vec{D}' \\ \vec{B}' \end{pmatrix}] + R(\xi) \vec{\nabla} \cdot \begin{pmatrix} \vec{J}_m \\ Z_0 \vec{J}_e \end{pmatrix}$$

$$\vec{\nabla} \cdot \begin{pmatrix} Z_0 \vec{D}' \\ \vec{B}' \end{pmatrix} = R(-\xi) \begin{pmatrix} Z_0 \rho_e \\ \rho_m \end{pmatrix}$$

Duality transformations

- Symmetrized Maxwell eqns with magnetic charge/current

$$\vec{\nabla} \cdot \begin{pmatrix} Z_0 \vec{D} \\ \vec{B} \end{pmatrix} = \begin{pmatrix} Z_0 \rho_e \\ \rho_m \end{pmatrix}, \quad \vec{\nabla} \times \begin{pmatrix} -\vec{E} \\ Z_0 \vec{H} \end{pmatrix} = \partial_t \begin{pmatrix} \vec{B} \\ Z_0 \vec{D} \end{pmatrix} + \vec{\nabla} \cdot \begin{pmatrix} \vec{J}_m \\ Z_0 \vec{J}_e \end{pmatrix}$$

- Consider the proper orthogonal transformation $R(\xi)$

$$R(\xi) = \begin{pmatrix} \cos \xi & \sin \xi \\ -\sin \xi & \cos \xi \end{pmatrix}, \quad R(\xi)^T = R(\xi)^{-1} = R(-\xi)$$

and the reflection χ

$$\chi = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \chi \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} b \\ a \end{pmatrix}, \quad \chi^T = \chi^{-1} = \chi, \quad R\chi R = \chi$$

$$\vec{\nabla} \times \begin{pmatrix} -\vec{E}' \\ Z_0 \vec{H}' \end{pmatrix} = \partial_t \begin{pmatrix} \vec{B}' \\ Z_0 \vec{D}' \end{pmatrix} + R(\xi) \vec{\nabla} \cdot \begin{pmatrix} \vec{J}_m \\ Z_0 \vec{J}_e \end{pmatrix}$$

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- We can choose ξ to make

$$\rho_m = 0, \quad \vec{J}_m = \vec{0}$$

and end up with the usual form of Maxwell eqns.

- There's also a choice that gives

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- Electric-vs-magnetic charge is really a convention within the generalized framework of electrodynamics
- the ratio of electric/magnetic charge changes by the same factor for all particles under a duality transformation

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Magnetic monopoles and quantum mechanics

In Proc. R. Soc. London Ser. A **133** (1931) 60, Dirac

Quantised Singularities in the Electromagnetic Field.

By P. A. M. DIRAC, F.R.S., St. John's College, Cambridge.

(Received May 29, 1931.)

§ 1. *Introduction.*

The steady progress of physics requires for its theoretical formulation a mathematics that gets continually more advanced. This is only natural and to be expected. What, however, was not expected by the scientific workers of the last century was the particular form that the line of advancement of the mathematics would take, namely, it was expected that the mathematics would get more and more complicated, but would rest on a permanent basis of axioms and definitions, while actually the modern physical developments have required a mathematics that continually shifts its foundations and gets more abstract. Non-euclidean geometry and non-commutative algebra, which were at one time considered to be purely fictions of the mind and pastimes for logical thinkers, have now been found to be very necessary for the description of general facts of the physical world. It seems likely that this process of increasing abstraction will continue in the future and that advance in physics is to be associated with a continual modification and generalisation of the axioms at the base of the mathematics rather than with a logical development of any one mathematical scheme on a fixed foundation.

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It thus appears that we must abandon the identification of the holes with protons and must find some other interpretation for them. Following Oppenheimer,[§] we can assume that in the world as we know it, *all*, and not merely nearly all, of the negative-energy states for electrons are occupied. A hole, if there were one, would be a new kind of particle, unknown to experimental physics, having the same mass and opposite charge to an electron. We may call such a particle an anti-electron. We should not expect to find any of them in nature, on account of their rapid rate of recombination with electrons, but if they could be produced experimentally in high vacuum they would be quite stable and amenable to observation. An encounter between two hard γ -rays (of energy at least half a million volts) could lead to the creation simultaneously of an electron and anti-electron, the probability of occurrence of this process being of the same order of magnitude as that of the collision of the two γ -rays on the assumption that they are spheres of the same size as classical

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† H. Weyl, 'Gruppentheorie und Quantenmechanik,' 2nd ed. p. 234 (1931).

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electrons. This probability is negligible, however, with the intensities of γ -rays at present available.

The protons on the above view are quite unconnected with electrons. Presumably the protons will have their own negative-energy states, all of which normally are occupied, an unoccupied one appearing as an anti-proton. Theory at present is quite unable to suggest a reason why there should be any differences between electrons and protons.

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Our theory thus allows isolated magnetic poles, but the strength of such poles must be quantised, the quantum μ_0 being connected with the electronic charge e by

$$hc/e\mu_0 = 2. \quad (9)$$

This equation is to be compared with (1). The theory also requires a quantisation of electric charge, since any charged particle moving in the field of a pole of strength μ_0 must have for its charge some integral multiple (positive or negative) of e , in order that wave functions describing the motion may exist.

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- but production cross sections not calculable through perturbation theory (like QED) due to strong coupling (large magnitude)

Search for magnetic monopoles

- Signal in a superconducting quantum interferometer device (SQUID) from material samples
- Funny tracks (large energy loss, non-helical path in B-field) in cosmic ray and collider experiments
- Other strange effects (different Cerenkov radiation, straight lightning, excess photon yield in cosmic ray showers, etc.)
- Magnitude of magnetic field (earth, galaxy) which would have been drained by accelerating them
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Experimental Upper Limit on the Magnetic Monopole Moment of Electrons, Protons, and Neutrons, Utilizing a Superconducting Quantum Interferometer

LORIN L. VANT-HULL

Ford Scientific Research Staff, Newport Beach, California 92663

(Received 15 April 1968)

A superconducting quantum interferometer (magnetic flux sensitivity $\sim 10^{-17}$ Wb) is shown to be capable of directly detecting magnetic charge, in particular Dirac's quantized magnetic monopoles (magnetic flux $= h/e = 4 \times 10^{-15}$ Wb). In samples free of Dirac monopoles, $\text{div} \mathbf{B}$ was determined to be less than 1.2×10^{-14} Wb/kg, while the magnetic charge of the neutron and of the proton (relative to the electron) were found to be less than 10^{-26} Dirac monopole per nucleon. The magnetic charge of the electron itself was measured to be less than 4×10^{-24} Dirac monopole.

MAGNETIC current will be detected by a superconducting quantum interferometer in a way analogous to the determination of electric current by a "clip-on ammeter." This type of ammeter responds to the solenoidal magnetic field generated by the electric current ($\text{curl} \mathbf{H} = \mathbf{j}_e + \partial \mathbf{D} / \partial t$). Similarly, the quantum interferometer should respond to the solenoidal electric field presumed generated by a magnetic current ($\text{curl} \mathbf{E} = -\mathbf{j}_m - \partial \mathbf{B} / \partial t$). Passage of a magnetic monopole charge (p) through the device at $t=0$ generates a voltage impulse $V(t) = p\delta(t)$, which acts to shift the quantum phase,

based on a particular assumed model of the monopole properties; i.e., its binding energy in magnetic or nonmagnetic media, its trajectory in electromagnetic fields, its ionization loss in a counter or nuclear emulsion, etc. Unfortunately, such experiments can do no more than show that particles meeting the requirements of the experiment were not found. A detection scheme which is independent of any model is thus called for, i.e., one utilizing a detector, such as a superconducting quantum interferometer, which requires only the presence or passage of a magnetic monopole.

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Magnetic monopoles and the survival of galactic magnetic fields

Michael S. Turner, E. N. Parker, and T. J. Bogdan

Astronomy and Astrophysics Center, The University of Chicago, Chicago, Illinois 60637

(Received 1 June 1982)

The most stringent, mass-independent limit on the flux of magnetic monopoles is based upon the survival of the galactic magnetic fields, the so-called "Parker limit": $F \lesssim 10^{-16} \text{ cm}^{-2} \text{sr}^{-1} \text{sec}^{-1}$. We reexamine this limit, taking into account the monopole's mass and velocity distribution, and the observed structure of the galactic magnetic field. We derive flux limits which depend upon the monopole's mass and velocity, and the strength, coherence length, and regeneration time of the galactic magnetic field. The largest monopole flux consistent with both the survival of the galactic magnetic field and the bounds from the mass density contributed by monopoles is $F \simeq 10^{-12} \text{ cm}^{-2} \text{sr}^{-1} \text{sec}^{-1}$, arising for monopoles of mass $\simeq 10^{19} \text{ GeV}$ with velocity $\simeq 3 \times 10^{-3} c$ which cluster with the Galaxy. An observed flux greater than this would have profound implications for our understanding of the galactic magnetic field, and we briefly explore some exotic possibilities. Of course, this bound is not applicable to a local source (e.g., the Sun, atmospheric cosmic-ray production, etc.).



Mass limit for Dirac-type magnetic monopoles

S. Graf, A. Schäfer, W. Greiner

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[https://doi.org/10.1016/0370-2693\(91\)90623-X](https://doi.org/10.1016/0370-2693(91)90623-X)

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Abstract

We calculate the contribution of virtual monopole-antimonopole pairs to the anomalous magnetic moment of the muon $(g-2)_\mu$. If strong coupling QED generates confinement of monopole-antimonopole pairs, as suggested by recent lattice calculations, most of the monopole search experiments would not have been capable of detecting monopoles. However, such confined systems contribute to radiative corrections. We conclude that the effective monopole mass has to be larger than 120 GeV.

Search for relativistic magnetic monopoles with IceCube

R. Abbasi *et al.* (IceCube Collaboration)

Phys. Rev. D **87**, 022001 – Published 18 January 2013

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ABSTRACT

We present the first results in the search for relativistic magnetic monopoles with the IceCube detector, a subsurface neutrino telescope located in the South Polar ice cap containing a volume of 1 km^3 . This analysis searches data taken on the partially completed detector during 2007 when roughly 0.2 km^3 of ice was instrumented. The lack of candidate events leads to an upper limit on the flux of relativistic magnetic monopoles of $\Phi_{90\% \text{C.L.}} \sim 3 \times 10^{-18} \text{ cm}^{-2} \text{ sr}^{-1} \text{ s}^{-1}$ for $\beta \geq 0.8$. This is a factor of 4 improvement over the previous best experimental flux limits up to a Lorentz boost γ below 10^7 . This result is then interpreted for a wide range of mass and kinetic energy values.

Search for non-relativistic magnetic monopoles with IceCube

IceCube Collaboration

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Slow monopole search limits – flux

Received: 14 February 2014 / Accepted: 11 June 2014 / Published online: 2 July 2014

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Abstract The IceCube Neutrino Observatory is a large Cherenkov detector instrumenting 1 km^3 of Antarctic ice. The detector can be used to search for signatures of particle physics beyond the Standard Model. Here, we describe the search for non-relativistic, magnetic monopoles as remnants of the *Grand Unified Theory* (GUT) era shortly after the Big Bang. Depending on the underlying gauge group these monopoles may catalyze the decay of nucleons via

the Rubakov–Callan effect with a cross section suggested to be in the range of 10^{-27} to 10^{-21} cm^2 . In IceCube, the Cherenkov light from nucleon decays along the monopole trajectory would produce a characteristic hit pattern. This paper presents the results of an analysis of first data taken from May 2011 until May 2012 with a dedicated slow-particle trigger for DeepCore, a subdetector of IceCube. A second analysis provides better sensitivity for the brightest non-relativistic monopoles using data taken from May 2009 until May 2010. In both analyses no monopole signal was observed. For catalysis cross sections of 10^{-22} (10^{-24}) cm^2 the flux of non-relativistic GUT monopoles is constrained up to a level of $\Phi_{90} \leq 10^{-18}$ (10^{-17}) $\text{cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$ at a 90 % confidence level, which is three orders of magnitude below the Parker bound. The limits assume a dominant decay of the proton into a positron and a neutral pion. These results

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Search for ultrarelativistic magnetic monopoles with the Pierre Auger observatory

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Ultrafast monopole search limits – flux

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PHYSICAL REVIEW D **94**, 082002 (2016)

We present a search for ultrarelativistic magnetic monopoles with the Pierre Auger observatory. Such particles, possibly a relic of phase transitions in the early Universe, would deposit a large amount of energy along their path through the atmosphere, comparable to that of ultrahigh-energy cosmic rays (UHECRs). The air-shower profile of a magnetic monopole can be effectively distinguished by the fluorescence detector from that of standard UHECRs. No candidate was found in the data collected between 2004 and 2012, with an expected background of less than 0.1 event from UHECRs. The corresponding 90% confidence level (C.L.) upper limits on the flux of ultrarelativistic magnetic monopoles range from $10^{-19}(\text{cm}^2 \text{ sr s})^{-1}$ for a Lorentz factor $\gamma = 10^9$ to $2.5 \times 10^{-21}(\text{cm}^2 \text{ sr s})^{-1}$ for $\gamma = 10^{12}$. These results—the first obtained with a UHECR detector—improve previously published limits by up to an order of magnitude.

DOI: [10.1103/PhysRevD.94.082002](https://doi.org/10.1103/PhysRevD.94.082002)

Search for slow magnetic monopoles with the NOvA detector on the surface

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Slow monopole search limits – flux



(Received 10 September 2020; accepted 22 December 2020; published 11 January 2021)

We report a search for a magnetic monopole component of the cosmic-ray flux in a 95-day exposure of the NOvA experiment's Far Detector, a 14 kt segmented liquid scintillator detector designed primarily to observe GeV-scale electron neutrinos. No events consistent with monopoles were observed, setting an upper limit on the flux of $2 \times 10^{-14} \text{ cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$ at 90% C.L. for monopole speed $6 \times 10^{-4} < \beta < 5 \times 10^{-3}$ and mass greater than $5 \times 10^8 \text{ GeV}$. Because of NOvA's small overburden of 3 meters-water equivalent, this constraint covers a previously unexplored low-mass region.

DOI: [10.1103/PhysRevD.103.012007](https://doi.org/10.1103/PhysRevD.103.012007)

I. INTRODUCTION

Magnetically charged particles were hypothesized by Dirac in 1931 [1] and are generically predicted by grand unified theories (GUTs) [2–4]. Although GUT-scale monopoles of $\sim 10^{17}$ – 10^{18} GeV are often assumed, recent theoretical work suggests possible masses as light as

$\sim 10^7 \text{ GeV}$ [2,5]. Searches over the past century for a monopole component of the cosmic ray flux have yet to find convincing evidence [6]. Slow-moving ($\beta < 0.01$) GUT-scale monopoles have been ruled out by underground experiments [7]. Weaker limits exist for slow monopoles in the range $10^5 \text{ GeV} < m < 10^{12} \text{ GeV}$ from mountaintop experiments [8]. In this paper, we focus on